INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH, MOHALI
MID SEMESTER EXAM
CHM101: BASIC INORGANIC CHEMISTRY

FULL MARKS: 25 **DURATION: 1 H**

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1. **a.** Draw the radial distribution function for 4s and 5d orbitals. Label the axis of your diagrams appropriately. *3*

b. Determine the order of increasing size for the following ions. *2*

K^+, Ca^{2+}, S^{2-}, P^{3-}

2. **a.** $R_2OBH_3 + R_2S \rightarrow$? Rationalize your answer. *2*

A sulfur-based ligand is given the choice of binding with $\underline{Pd^{2+}}$ and $\underline{Pd^{4+}}$. In which case the binding will be stronger? *2*

b. Define polarizability. Describe how larger size of an anion can form a more covalent bond with Al^{3+}. *2*

c. Applying Slater's rule, determine the Z_{eff} for Cr and F. *4*

3. Explain the trend of solubility for silver halide salts. Silver iodides are completely insoluble in water, even after the addition of ammonia, while silver chloride is moderately (in presence of ammonia) soluble. Silver fluoride is completely soluble in water. *2*

4. **a.** Draw the molecular orbital energy level diagram of N_2 and determine the bond order of nitrogen. Discuss the ground state magnetic property of B_2. *3+1*

b. Draw pictures of π^* molecular orbital of CO. Show pictorially how backbonding happens from an electron rich metal center. *2*

c. Draw the molecular orbital diagram of HCl. *2*